

1. System Overview & Architecture

Enterprise Financial Transaction Core for Hotel, Restaurant & Asset Operations

1.1 About the HORECA Financial ERP

The HORECA Financial Management System is a specialized enterprise resource planning (ERP) platform designed specifically for hospitality operations (Hotels, Restaurants, Catering, and Asset Management). It centralizes financial transactions from four core operational modules into an immutable, double-entry General Ledger (GL) powered by PostgreSQL standards.

Connected Subsystem	Data Stream Generated	General Ledger Integration Rule
Hotel PMS	Guest folios, lodging revenue, mini-bar, VAT 12%, service charge pool (85%)	Debits Cash Float (1010) / City Ledger (1210); Credits Lodging Revenue (4010) & Tax
Restaurant POS	Dining room food orders, bar beverage sales, till cash drops, credit swipes	Debits Till Float (1020) / Card Settlement (1040); Credits Food Sales (4020) & Bar (4030)
HRMS Payroll	Bi-monthly staff wages, GM salaries, SSS/PhilHealth/HDMF, withholding tax	Debits Salaries Expense (5110/5120); Credits Accrued Payroll (2030) & Withholding (2110)
Supply Chain	Purveyor meat/produce deliveries, dry goods POs, vendor 3-way match bills	Debits COGS Inventory (5010/5020); Credits Trade Accounts Payable (2010)

1.2 Key Security & Compliance Tenets

- ½ Default Data Masking: Sensitive fields (tax IDs, card numbers, bank routing, executive compensation) are masked upon login by default.
- ½ Two-Tier Maker-Checker Governance: Standard Admins queue proposals; only Super Admins can authorize cross-module commits.
- ½ 2-Factor Google OTP Verification: Every login triggers a dynamic 6-digit OTP sent to the user's registered Google account (Gmail).
- ½ 15-Minute Session Inactivity Timeout: Automatic session termination after 15 minutes of inactivity to protect workstations.
- ½ Immutable Audit Trail: Every state transition captures user ID, client IP, timestamp, and full before/after JSON diffs.

2. Authentication, Google OTP & User Roles

Security Protocols, Credential Management & Two-Tier Approvals

2.1 Default System Accounts & Credentials

User Role	System Login Email	Linked Google Account (Gmail)	Password	Permissions
Super Administrator	Lorenz@horeca.com	gaumlorenz@gmail.com	230117492	Full System Master Access: Approve all queues, unmask data password control, create/manage admin users, post GL journals directly.
Standard Administrator	Renz@horeca.com	grave3116@gmail.com	#Ga2004	Operational Access: Draft invoices, submit collection receipts, request disbursements, propose budget allocations (Requires Super Admin Approval).

2.2 Google Account OTP Verification Flow

When logging into the system, the user enters their System Email and Password. Upon successful initial authentication, the system dispatches a secure 6-digit One-Time Password (OTP) to their linked Google account (Gmail). The user must enter the OTP within 5 minutes to complete authentication. Each session timeout re-triggers this OTP requirement.

2.3 Super Admin User Management Settings

- Super Admin navigates to the 'User Management' setting in the sidebar.
- To add a new Standard Admin, fill in: Full Name, System Email (@horeca.com), and the staff member's official Google Account (Gmail).
- Upon saving, the user is registered in the Authorized Users table and can log in using their credentials + Gmail OTP.
- Super Admin can instantly suspend, activate, or reset OTP tokens for any user from the centralized table.

2.4 Two-Tier Approval Governance (Maker-Checker)

Whenever a Standard Admin performs an action (such as creating an AP vendor bill, logging cash receipts, requesting salary disbursements, or adjusting budget limits), the transaction is placed into the 'Governance & Approval Requests Queue' with status 'Pending'. The Super Admin inspects the payload diff and cross-module impact. Upon Super Admin approval, the transaction is atomically committed across all modules and reflected immediately in the Standard Admin's view.

3. General Ledger Transaction Core

Master Chart of Accounts, Double-Entry Balancing & Transaction History

3.1 General Ledger Capabilities

The General Ledger is the central financial hub of the ERP. It provides real-time verification that Total Debits strictly equal Total Credits (Zero Variance). It contains the full Hospitality Chart of Accounts (COA) spanning Assets (1000s), Liabilities (2000s), Equity (3000s), Revenue (4000s), and Expenses (5000s).

GL View Tab	Purpose & Operational Instructions
Master GL Register	Chronological audit table of all posted debit and credit lines across Hotel PMS, POS, Payroll, and Supply Chain. Filterable by module and account.
Transaction History	Grouped batch vouchers (e.g., JV-2026-PMS-01, INV-8821, COL-101) showing timestamp, source module, balancing verification, posted by, and approval status with expandable line items.
Journal Entry Builder	Interactive double-entry builder with preset hospitality templates (Night Audit, Dinner Service, Payroll, Purveyor Invoice). Enforces mathematical debit=credit balance before submission.
Hospitality Chart of Accounts	Complete 22+ standard account dictionary with classification, normal balance (Debit/Credit), and financial statement mapping.
PostgreSQL DDL & Schema	Complete SQL migration scripts, table schemas (coa, journal_entries, subsystems), constraints, and Docker deployment seeds.
Data Dictionary	Technical field specifications, PostgreSQL data types, foreign keys, and audit definitions.

3.2 Using the Transaction History View

- Navigate to 'General Ledger' in the sidebar and select the 'Transaction History' sub-tab.
- Review summarized transaction batches grouped by voucher reference (JV/INV/COL/DISB).
- Inspect balancing status: Green indicates 100% Balanced (Zero Variance); Red indicates an out-of-balance transaction.
- Click 'View Journal Breakdown' on any transaction row to expand full debit and credit line details.
- Use 'Export to Excel' or 'Export to PDF' to download complete historical audit records.

4. Core Operational Financial Modules

AP/AR Invoices, Collections, Disbursements & Budget Linear Regression

Module	Core Features & Hospitality Procedures
Accounts Payable & Accounts Receivable (AP/AR)	4-Bucket Aging Schedule: Categorizes trade liabilities and guest receivables into 0-30, 31-60, 61-90, and 90+ days. 3-Way Match Verification: Validates Purchase Order (PO) Delivery Receipt, Supplier Invoice, and Warehouse Receipt. Trade Settlement: Direct settlement adjusts cash vaults, clears invoices, and posts balancing GL journals.
Collection Management	Multi-Tender Tracking: Records cash drawer drops, credit card merchant settlements, and GCash/Maya e-wallets. Shift-End Drawer Reconciliation: Compares expected register intake against physical cash counts to detect variances. Target Vault Routing: Routes funds directly to Operating Bank Account or Front Desk Petty Cash.
Disbursement Control	4-Stage Approval Protocol: Draft Verified GM Approved Bank Release 2-Tier Signature Verification: Requires Finance Officer review and General Manager authorization. Creditable Withholding Tax (EWT): Computes 2% expanded withholding for BIR Form 2307 compliance.
Budget Management & Linear Regression	Ordinary Least Squares (OLS) Regression: Forecasts future departmental expenses using mathematical trend lines ($y = mx + b$). Dynamic Scenario Modeling: Simulates expense projections based on inflation rates (1%-15%) and hotel occupancy (30%-100%). CapEx vs OpEx Allocation: Real-time spend vs limit monitoring with risk threshold alerts (>90% Critical Risk).

4.2 How to Run Budget Linear Regression Forecasting

- Navigate to 'Budget Management' in the sidebar.
- The 'Linear Regression Expense Forecasting Engine' automatically computes historical slope (m) and intercept (b).
- Adjust the 'Projected Inflation Rate' slider to test economic cost fluctuations.
- Adjust the 'Projected Hotel Occupancy Rate' slider to evaluate variable guest service overheads.
- Review the Recharts visual projection diagram and forecasted monthly department targets.
- Click 'Export Budget Report' to generate formal Excel or PDF forecasts for executive board review.

5. Cash Treasury, Tax, Audit & Security Settings

Liquidity Vaults, BIR Tax Compliance, Masking Password & Session Controls

Module / Function	Standard Operating Procedure & Security Rules
Cash & Treasury Management	🔗 Dual Liquidity Vaults: Monitors Primary Operating Bank Account (BDO/BPI) and Physical Petty Cash Float. 🔗 Bank Statement Reconciliation: Side-by-side comparison of internal GL cash lines against imported commercial bank statements. 🔗 Liquidity Reserve Alert: Warns management if available bank balance drops threshold.
Tax Management & BIR Filings	🔗 Value Added Tax (VAT 12%): Monthly tax base calculation and return schedule for BIR Form 2550M. 🔗 Expanded Withholding Tax (EWT 2% / 10%): Vendor tax deductions and BIR Form 2307 certificate generation. 🔗 Compensation Withholding Tax: Payroll tax deduction tracking for monthly BIR Form 1601-C filing.
Super Admin Unmask Password Control	🔗 Default Unmask Password: #S230117482 🔗 Super Admin Configuration: Super Admin can change the unmask password anytime in the Security Settings tab. 🔗 Unmasking Data: Clicking 'Unmask Sensitive Data' requires entering this authorized password.
15-Minute Session Inactivity Timeout	🔗 Active Countdown: The system displays a live session timer in the sidebar. 🔗 Inactivity Reset: Every mouse movement or keystroke refreshes the active session. 🔗 Auto-Lock: After 15 minutes of continuous inactivity, the session automatically terminates and requires re-login with Gmail OTP.
Collapsible Icon-Only Sidebar	🔗 Toggle Button: Click the collapse icon on the top-left sidebar header to minimize the navigation rail to icon-only mode (w-20). 🔗 Tooltips: Hovering over icons displays clean module tooltips.

5.2 Quick Reference Checklist for Daily Financial Operations

- ' Morning: Verify Operating Bank & Petty Float balances under 'Cash Management'.
- ' Mid-Day: Process trade purveyor bills in 'AP/AR' with 3-Way Match validation.
- ' Shift-End: Perform drawer drop count reconciliation under 'Collections'.
- ' Night Audit: Post balanced hotel lodging & restaurant dinner journals in 'General Ledger'.
- ' Executive Review: Inspect 'Transaction History' and 'FP&A Financial Reporting' statements.
- ' Month-End: Execute 'Budget Linear Regression Forecast' and file BIR statutory tax schedule.